When Models Surprise Us

Phase Transitions, Grokking, and the Test-Time Compute Revolution

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LLM Lab — Executive Education Program

2026

The Unifying Thread

"The dynamics of learning in LLMs violate our deepest intuitions."

A. Phase Transitions

Scale a model 10× and it suddenly writes poetry.

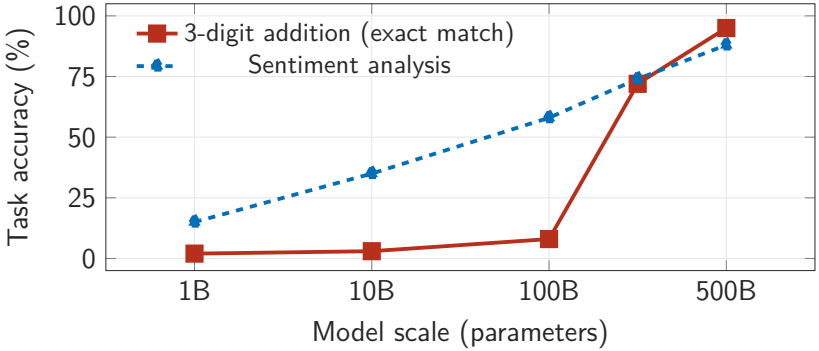
B. Grokking

A model memorises, stalls — then *understands*.

C. Test-Time Compute

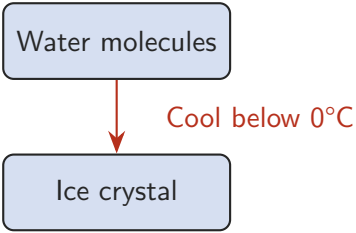
Let a small model think longer and it beats a giant.

Emergent Abilities: The Soufflé Problem

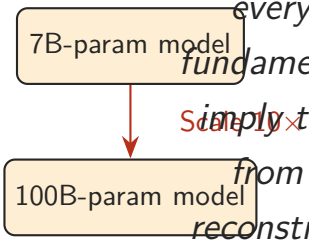


Imagine scaling a recipe: double the flour, double the output. Now imagine that at exactly 10× flour, the cake spontaneously becomes a soufflé. That is what Wei et al. observed in LLMs. [2]

“More Is Different” — From Physics to AI



Same H_2O molecules, new macroscopic property



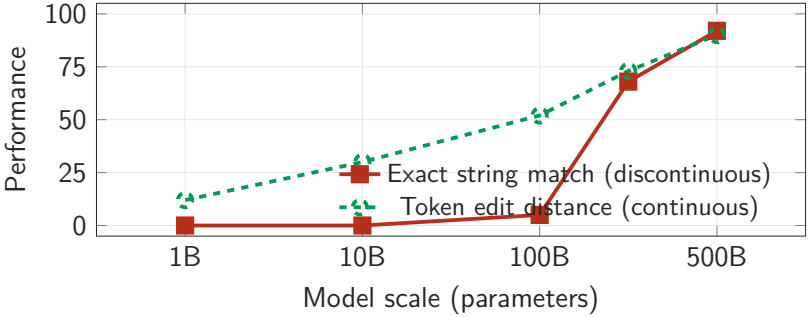
Same architecture, new cognitive capability

“The ability to reduce everything to simple fundamental laws does not imply the ability to start from those laws and reconstruct the universe.”

— P.W. Anderson, 1972 [1]

Nobel laureate. The same principle that explains why ice is rigid may explain why GPT-4 can write code

Or Is It a Mirage?



Schaeffer et al. (NeurIPS 2023, Outstanding Paper): 92% of claimed emergent abilities in BIG-Bench appeared only under **discontinuous metrics** (exact match, multiple-choice grade). Switch to a continuous metric $\Rightarrow$ smooth, predictable improvement.

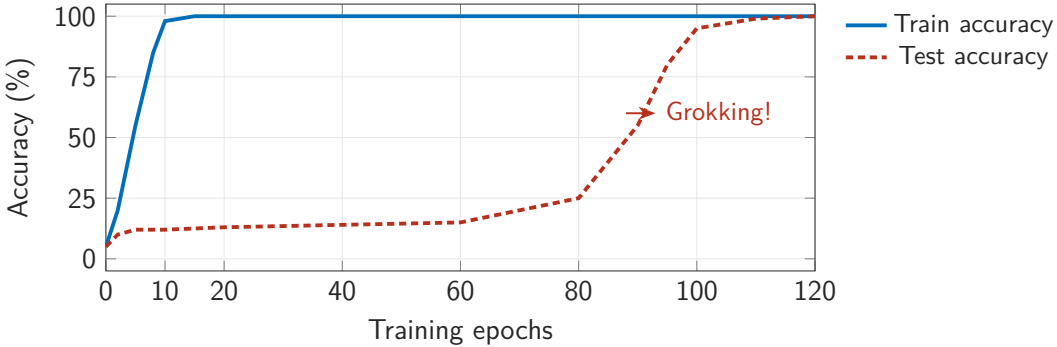
So capabilities can appear suddenly at scale.

But what if a model already *has* the data it needs

—

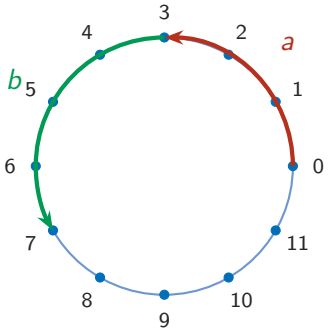
and still takes forever to understand it?

Grokking: Memorisation → Stall → Understanding



A model memorises every training example, loss goes flat, and *long after you would have stopped training* — generalisation appears. [5]

The Shocking Explanation: Fourier Circuits



$(a + b) \bmod 12$

What the network learns:

1. Embed inputs as $\sin(\omega a), \cos(\omega a)$
2. Compose rotations via trig identities:
   $$\cos \omega(a+b) = \cos \omega a \cos \omega b - \sin \omega a \sin \omega b$$
3. Read off $(a+b) \bmod p$

The network independently discovers discrete Fourier analysis. [6]

Grokking Is Not a Toy Phenomenon

Grokking at 7B scale

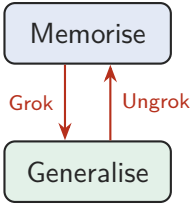
Li et al. (2025): "local grokking" during standard pretraining of a 7B-parameter model. Different domains (maths, code, commonsense) grok *asynchronously*. Training loss has converged; generalisation improves silently underneath. [8]

Business implication:

A model's capabilities may still be *emerging* even after training loss has plateaued and the budget is spent.

Ungrokking

Varma et al. (ICLR 2025): grokking can run *in reverse*. A model that has achieved perfect generalisation can regress back to memorisation under changed conditions. Generalisation is not permanent. [7]



Phase transitions say: *scale changes what a model can do.*

Grokking says: *time changes what a model understands.*

What if we give the model more time — *at inference*?

The New Paradigm: Let Models Think Longer

Old paradigm
Make models *bigger*

2024–2025 →

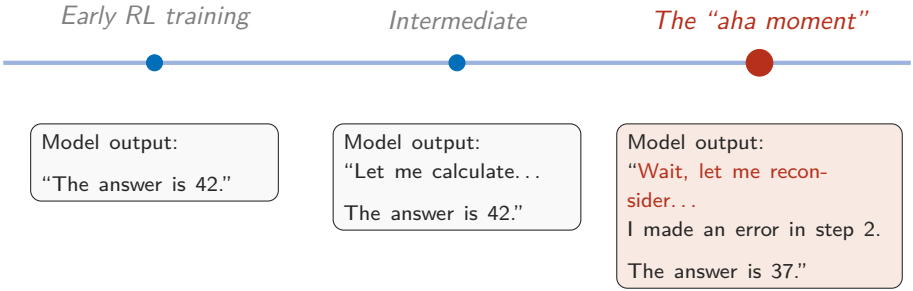
New paradigm
Let models *think longer*

GPT-3 → GPT-4
175B → 1.8T params

o1, DeepSeek-R1, s1
Same model, more thinking

Key result (Snell et al., ICLR 2025 Oral):
A smaller model using optimal test-time compute can outperform a 14× larger model in a FLOPs-matched comparison. [11]

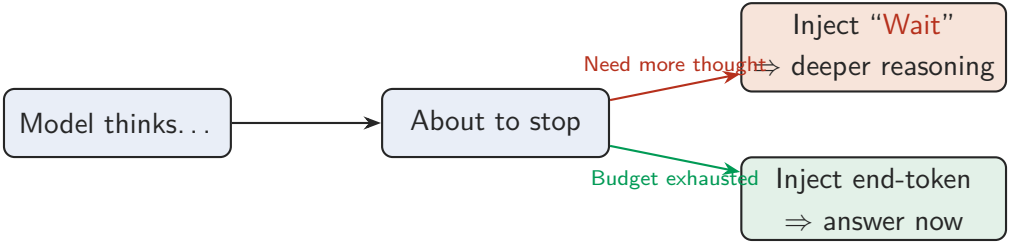
The “Aha Moment”: Reasoning from Pure Trial and Error



DeepSeek-R1 (2025): trained with *pure* reinforcement learning — no human demonstrations of reasoning. The model independently discovered that pausing to self-correct improves its answers. Nobody taught it to think step by step. [9]

A model trained only to maximise a reward signal independently discovers that reflection is useful. Is that metacognition — or just anthropomorphism?

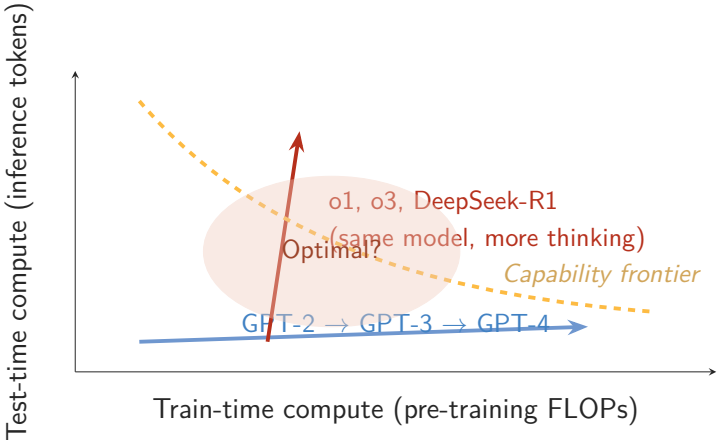
Budget Forcing: Controlling How Long a Model Thinks



s1-32B (Muennighoff et al., 2025): fine-tuned on only **1,000 examples**, beats OpenAI o1-preview on competition maths by up to 27% using budget forcing at test time. [10]

More thinking ⇒ better answers. The relationship is remarkably clean — a new scaling axis that is *cheaper* than training a larger model.

Two Axes of Scaling



The capability frontier now has *two dimensions*. The strategic question for any organisation deploying LLMs: where on this frontier should you invest? [11]

Questions to Take Home

1. If we cannot predict what the next model will be capable of, how should organisations plan for AI adoption?
   (Phase transitions mean capability forecasting is unreliable.)

2. Grokking suggests that understanding can emerge *long after* the data is memorised. What does this mean for AI safety?
   (A model might develop dangerous capabilities after we think training is "done.")

3. Test-time compute trades latency for accuracy. Which of your business decisions warrant a model that "thinks longer"?
   (Not all queries are equal — when is 30 seconds of thinking worth the cost?)

Further Reading

Topic	Start here
Phase transitions	Wei et al., "Emergent Abilities of LLMs" (TMLR 2022) [2]
	Schaeffer et al., "Are Emergent Abilities a Mirage?" (NeurIPS 2023) [3]
Grokking	Power et al., "Grokking" (ICLR 2022 Workshop) [5] Nanda et al., "Progress Measures for Grokking" (ICLR 2023) [6]
Test-time compute	DeepSeek-AI, "DeepSeek-R1" (2025) [9]
	Snell et al., "Scaling LLM Test-Time Compute" (ICLR 2025) [11]

Full reading list with accessible summaries in the companion document.

The dynamics of learning in LLMs violate our deepest intuitions.

Plan accordingly.

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P.W. Anderson.

More Is Different.

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J. Wei, Y. Tay, R. Bommasani, C. Raffel, B. Zoph, S. Borgeaud et al.

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Transactions on Machine Learning Research (TMLR), 2022.

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R. Schaeffer, B. Miranda, S. Koyejo.

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arXiv:2503.05788, 2025.

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Progress Measures for Grokking via Mechanistic Interpretability.

ICLR, 2023. Oral.

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V. Varma, R. Shah, Z. Kenton, J. Kramar, R. Kumar.
Explaining Grokking Through Circuit Efficiency.
ICLR, 2025.

Z. Li, C. Fan, T. Zhou.
Where to Find Grokking in LLM Pretraining?
arXiv:2506.21551, 2025.

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DeepSeek-AI, D. Guo, D. Yang et al.

DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning.

arXiv:2501.12948, 2025. Published in *Nature*.

N. Muennighoff, Z. Yang, W. Shi, X.L. Li, L. Fei-Fei et al.

s1: Simple Test-Time Scaling.

arXiv:2501.19393, 2025.

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C. Snell, J. Lee, K. Xu, A. Kumar.

Scaling LLM Test-Time Compute Optimally Can Be More Effective Than Scaling Model Parameters.

ICLR, 2025. Oral.